

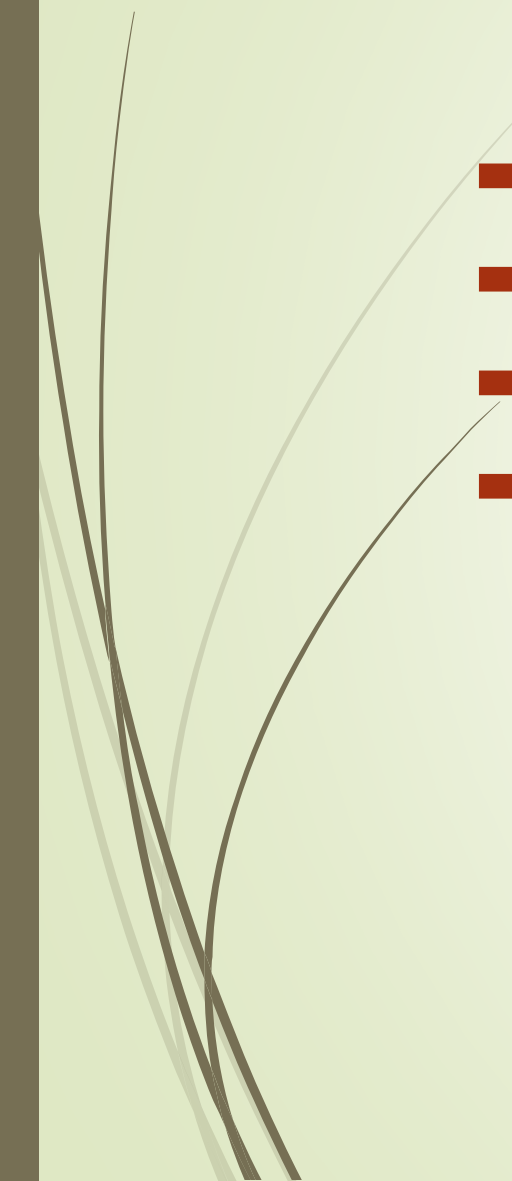


Chapter 5

Discriminant Analysis



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§ 5.1 Introduction

➤ Discriminant analysis examples:

- 1. Solvent(有偿付能力) and insolvent property liability insurance company:

Measured variables: total assets, stock and bond value, stock and bond market value, loss expenditure, surplus, signed premium (签订的保费金额).

- 2. Good and bad credit risk:

Measured variables: income, age, career, numbers of credit cards, family size.

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- Discriminant analysis is to solve that which group a new observed sample will be divided into based on the condition of known grouping.
 - $\mathbf{x} = (x_1, x_2, \dots, x_p)'$: p dimensional index values of the samples in every group, it is from a p dimensional population distribution. We attempt to judge which group a new sample $\mathbf{x}$ is from based on population distributions or distribution characteristics.
 - Three common discriminant methods are introduced in this chapter:
 - I- distance discriminant;
 - II- Bayes discriminant ;
 - III- Fisher discriminant.



§ 5.2 Distance Discriminant

- § 5.2.1 two groups distance discriminant
- § 5.2.2 multiple groups distance discriminant

§ 5.2.1 two group distance discriminant

- Suppose the means of group π_1 and π_2 are respectively $\boldsymbol{\mu}_1$ and $\boldsymbol{\mu}_2$ and the covariance matrices are $\boldsymbol{\Sigma}_1$ and $\boldsymbol{\Sigma}_2$ ($\boldsymbol{\Sigma}_1, \boldsymbol{\Sigma}_2 > 0$). $\mathbf{x}$ is a new sample (p -dimension). Now try to judge which group $\mathbf{x}$ is from
- 1. $\boldsymbol{\Sigma}_1 = \boldsymbol{\Sigma}_2 = \boldsymbol{\Sigma}$
- 2. $\boldsymbol{\Sigma}_1 \neq \boldsymbol{\Sigma}_2$



1. $\Sigma_1 = \Sigma_2 = \Sigma$

➡ **Discriminant Rule:**
$$\begin{cases} \mathbf{x} \in \pi_1, & \text{if } d^2(\mathbf{x}, \pi_1) \leq d^2(\mathbf{x}, \pi_2) \\ \mathbf{x} \in \pi_2, & \text{if } d^2(\mathbf{x}, \pi_1) > d^2(\mathbf{x}, \pi_2) \end{cases}$$

➡ Let

$$\begin{aligned} & d^2(\mathbf{x}, \pi_1) - d^2(\mathbf{x}, \pi_2) \\ &= (\mathbf{x} - \mu_1)' \Sigma^{-1} (\mathbf{x} - \mu_1) - (\mathbf{x} - \mu_2)' \Sigma^{-1} (\mathbf{x} - \mu_2) \\ &= -2\mathbf{a}'(\mathbf{x} - \bar{\mu}) \end{aligned} \tag{5.2.2}$$

Where $\bar{\mu} = \frac{1}{2}(\mu_1 + \mu_2)$, $\mathbf{a} = \Sigma^{-1}(\mu_1 - \mu_2)$. Let $W(\mathbf{x}) = \mathbf{a}'(\mathbf{x} - \bar{\mu})$. Then

the above rule can be simplified as:
$$\begin{cases} \mathbf{x} \in \pi_1, & \text{if } W(\mathbf{x}) \geq 0 \\ \mathbf{x} \in \pi_2, & \text{if } W(\mathbf{x}) < 0 \end{cases} \tag{5.2.3}$$

➡ $W(\mathbf{x})$ is called **(Linear) Discriminant Function** (判别函数) of two group distance discriminant and $\mathbf{a}$ are called **Discriminant Coefficient** (判别系数).

Misclassification Probability

- Misclassification probability:

$$P(2|1) = P(W(\mathbf{x}) < 0 | \mathbf{x} \in \pi_1)$$

$$P(1|2) = P(W(\mathbf{x}) \geq 0 | \mathbf{x} \in \pi_2)$$

- Suppose $\pi_1 \sim N_p(\boldsymbol{\mu}_1, \boldsymbol{\Sigma})$, $\pi_2 \sim N_p(\boldsymbol{\mu}_2, \boldsymbol{\Sigma})$, then $x \in \pi_1$, $a = \boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)$,
 $W(x) = a'(x - \bar{\boldsymbol{\mu}}) \sim N(\frac{1}{2}a'(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2), a'\boldsymbol{\Sigma}a)$ $W(x) \sim N(\frac{1}{2}\Delta^2, \Delta^2)$

Exercise 1 $P(2|1) = P(1|2) = \Phi\left(-\frac{\Delta}{2}\right)$ (5.2.4)

where $\Delta = \sqrt{(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)'\boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)}$ are Mahalanobis distance between two groups.

Misclassification Probability

➤ **Exercise 1: Prove that** $P(2|1) = P(1|2) = \Phi\left(-\frac{\Delta}{2}\right)$.

➤ **Proof:** Suppose $\pi_1 \sim N_p(\mu_1, \Sigma)$, $\pi_2 \sim N_p(\mu_2, \Sigma)$. If $x \in \pi_2$, we have

$$W(x) = a'(x - \bar{\mu}) \sim N\left(\frac{1}{2}a'(\mu_2 - \mu_1), a'\Sigma a\right) \text{ and } W(x) \sim N\left(-\frac{1}{2}\Delta^2, \Delta^2\right),$$

where $\Delta = \sqrt{(\mu_1 - \mu_2)'\Sigma^{-1}(\mu_1 - \mu_2)}$. It follows that

$$P(1|2) = P(w(x) > 0) = P\left[\frac{w(x) - (-0.5\Delta^2)}{\Delta} > \frac{1}{2}\Delta\right] = \Phi\left(-\frac{\Delta}{2}\right).$$

If $x \in \pi_1$, $W(x) = a'(x - \bar{\mu}) \sim N\left(\frac{1}{2}a'(\mu_1 - \mu_2), a'\Sigma a\right)$, $W(x) \sim N\left(\frac{1}{2}\Delta^2, \Delta^2\right)$. Thus,

$$P(2|1) = P(w(x) < 0) = P\left[\frac{w(x) - 0.5\Delta^2}{\Delta} < -\frac{1}{2}\Delta\right] = \Phi\left(-\frac{\Delta}{2}\right),$$

➤ Obviously, the bigger Δ of two normal groups is, the smaller misclassification probabilities are. The discriminant result is more effective. On the contrary, when Δ becomes smaller, misclassification probabilities will increase. Discriminant analysis is meaningless when Δ approaches zero.

Define whether the distance between groups are too close

- How to define two groups is so close that discriminant analysis of them has no practical significance. We can consider the following hypothesis.
- Hypothesis test: $H_0: \mu_1 = \mu_2$, $H_1: \mu_1 \neq \mu_2$. If the test result is to accept H_0 , then two means are not significantly different, discriminant analysis is vain in this distribution;
- If the hypothesis test rejects H_0 , although there are significant differences between the means of the two groups, these differences may not be large enough for effective discriminant analysis (meaning that discriminant analysis may not be practically meaningful), Therefore, whether the misclassification probabilities exceed a rational level is another standard to concern.

- **Example 5.2.1** Suppose $p=1$, the distributions of π_1 and π_2 are respectively $N(\mu_1, \sigma^2)$ and $N(\mu_2, \sigma^2)$, μ_1, μ_2, σ^2 are known, $\mu_1 < \mu_2$. Then discriminant coefficient $a = (\mu_1 - \mu_2) / \sigma^2 < 0$,

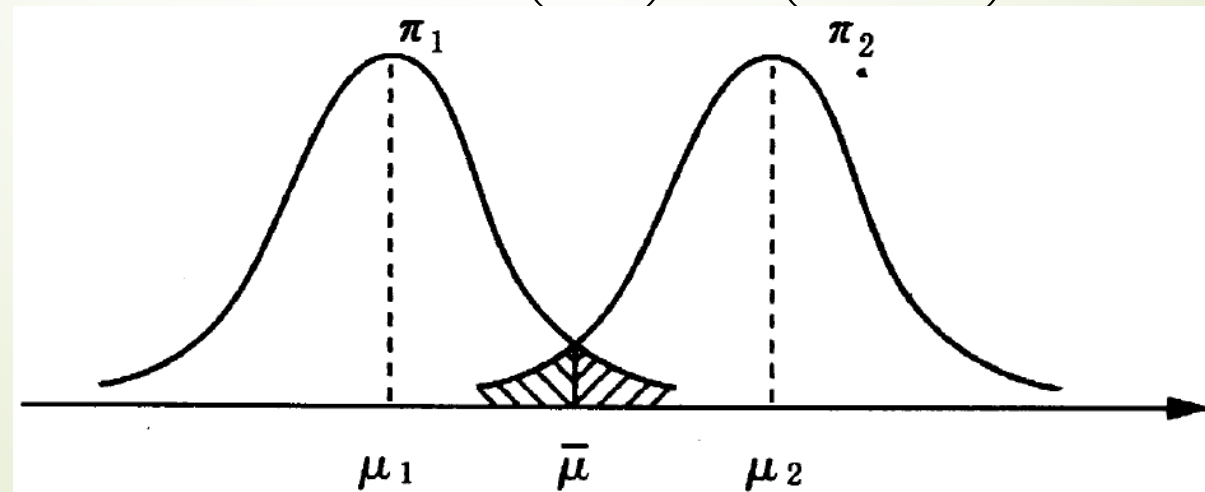
Discrimination Function: $W(x) = a(x - \bar{\mu})$

Discrimination Rule:
$$\begin{cases} x \in \pi_1, & x \leq \bar{\mu} \\ x \in \pi_2, & x > \bar{\mu} \end{cases} \quad \begin{cases} \mathbf{x} \in \pi_1, & \text{if } W(\mathbf{x}) \geq 0 \\ \mathbf{x} \in \pi_2, & \text{if } W(\mathbf{x}) < 0 \end{cases}$$

Misclassification Probability: $\Delta = \sqrt{(\mu_1 - \mu_2)' \Sigma^{-1} (\mu_1 - \mu_2)}$

$$P(2|1) = P(1|2) = \Phi\left(-\frac{\Delta}{2}\right) = \Phi\left(\frac{\mu_1 - \mu_2}{2\sigma}\right)$$

Figure:



Real application for distance discriminant

- Suppose $\mathbf{x}_{11}, \mathbf{x}_{12}, \dots, \mathbf{x}_{1n_1}$ are samples from group π_1 , $\mathbf{x}_{21}, \mathbf{x}_{22}, \dots, \mathbf{x}_{2n_2}$ are from group π_2 , $n_1 + n_2 - 2 \geq p$. Then the unbiased estimate of $\boldsymbol{\mu}_1$ and $\boldsymbol{\mu}_2$ are respectively:

$$\bar{\mathbf{x}}_1 = \frac{1}{n_1} \sum_{j=1}^{n_1} \mathbf{x}_{1j} \quad , \quad \bar{\mathbf{x}}_2 = \frac{1}{n_2} \sum_{j=1}^{n_2} \mathbf{x}_{2j}$$

One joint unbiased estimate of $\boldsymbol{\Sigma}$ is:

$$\mathbf{S}_p = \frac{1}{n_1 + n_2 - 2} (\mathbf{A}_1 + \mathbf{A}_2)$$

where $\mathbf{A}_i = \sum_{j=1}^{n_i} (\mathbf{x}_{ij} - \bar{\mathbf{x}}_i)(\mathbf{x}_{ij} - \bar{\mathbf{x}}_i)'$, $i = 1, 2$

- Discrimination function of estimate is :

$$\hat{W}(\mathbf{x}) = \hat{\mathbf{a}}'(\mathbf{x} - \bar{\mathbf{x}}) \quad (5.2.5)$$

where $\bar{\mathbf{x}} = \frac{1}{2}(\bar{\mathbf{x}}_1 + \bar{\mathbf{x}}_2)$, $\hat{\mathbf{a}} = \mathbf{S}_p^{-1}(\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2)$ The discrimination rule:

$$\begin{cases} \mathbf{x} \in \pi_1, & \hat{W}(\mathbf{x}) \geq 0 \\ \mathbf{x} \in \pi_2, & \hat{W}(\mathbf{x}) < 0 \end{cases} \quad (5.2.6)$$

- If π_1 and π_2 are **normal groups(正态组)**, then misclassification probabilities $P(2|1)$ and $P(1|2)$ can be estimated as:

$$\hat{P}(2|1) = \hat{P}(1|2) = \Phi\left(-\frac{\hat{\Delta}}{2}\right)$$

where $\hat{\Delta} = \sqrt{(\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2)' \mathbf{S}_p^{-1} (\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2)}$.

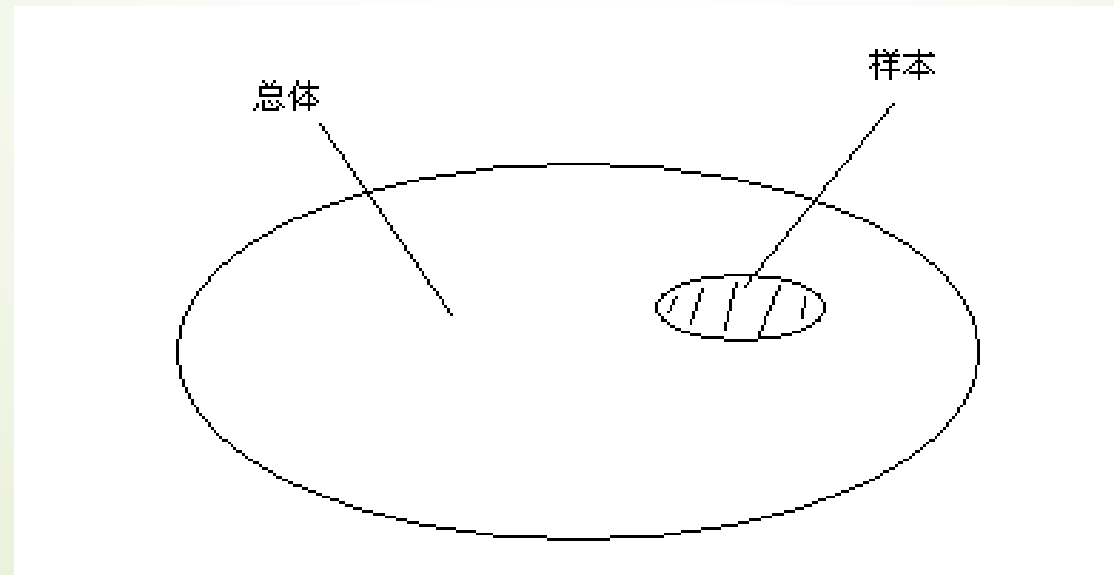
Non-parameter estimate of misclassification probability

- If the group cannot be supposed as normal, then $P(2|1)$ and $P(1|2)$ can be estimated by the sample misclassification ratio, usually obtained by the following three methods:
- (1) **Frequency approximation.** Suppose $n(2|1)$ is the number of samples belonging to π_1 but being misclassified to π_2 and $n(1|2)$ the number of samples belonging to π_2 but being misclassified to π_1 . Then $P(2|1)$ and $P(1|2)$ can be estimated as :

$$\hat{P}(2|1) = \frac{n(2|1)}{n_1}, \quad \hat{P}(1|2) = \frac{n(1|2)}{n_2} \quad (5.2.7)$$

- This method is simple, visualized and easy to calculate. It is a pity that the estimate is usually low unless n_1 and n_2 are both extremely big.

- The reason for the appearance of this kind of optimistic estimate is that the sample data applied to construct the discrimination function are also used to evaluate the same function. Obviously, the evaluation result is beneficial to discrimination function. In fact, for the estimate of misclassification probability, when the discrimination function is evaluated, the sample data used to construct it cannot represent the population effectively.



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- (2) Divide the whole sample into two parts. One portion is training sample(训练样本) used for establishing discrimination function and another portion is verification sample(验证样本) applied to evaluate discrimination function. The misclassification probability is estimated by using the probability of verification sample being misclassified. In this way, the estimate is unbiased.
 - (3) Cross Validation Method(交叉验证法 or 刀切法) . This method not only can avoid irrationally reusing sample data to both construct and evaluate discrimination function, but also almost get rid of sample information loss.

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- Procedures: choose $\mathbf{x}_{1j}$ from group π_1 and construct discriminant function with the rest $n_1 - 1$ observed values from group π_1 and n_2 observed values of group π_2 , then discriminate $\mathbf{x}_{1j}$, $j = 1, 2, \dots, n_1$. In the same way, choose $\mathbf{x}_{2j}$ from group π_2 and construct discriminant function with the rest $n_2 - 1$ observed values from group π_2 and n_1 observed values of group π_1 then discriminate $\mathbf{x}_{2j}$, $j = 1, 2, \dots, n_2$. Suppose $n^*(2|1)$ is the number of the samples belonging to π_1 but misclassified to π_2 , $n^*(1|2)$ represents the converse number. Then the estimates of misclassification probabilities $P(2|1)$ and $P(1|2)$ are

$$\hat{P}(2|1) = \frac{n^*(2|1)}{n_1}, \quad \hat{P}(1|2) = \frac{n^*(1|2)}{n_2} \quad (5.2.8)$$

They are approximately unbiased estimates.

2. $\Sigma_1 \neq \Sigma_2$

➤ Discrimination rule:

$$\begin{cases} \mathbf{x} \in \pi_1, & d^2(\mathbf{x}, \pi_1) \leq d^2(\mathbf{x}, \pi_2) \\ \mathbf{x} \in \pi_2, & d^2(\mathbf{x}, \pi_1) > d^2(\mathbf{x}, \pi_2) \end{cases} \quad (5.2.9)$$

➤ Another form of discrimination function:

$$\begin{aligned} W(\mathbf{x}) &= d^2(\mathbf{x}, \pi_1) - d^2(\mathbf{x}, \pi_2) \\ &= (\mathbf{x} - \boldsymbol{\mu}_1)' \boldsymbol{\Sigma}_1^{-1} (\mathbf{x} - \boldsymbol{\mu}_1) - (\mathbf{x} - \boldsymbol{\mu}_2)' \boldsymbol{\Sigma}_2^{-1} (\mathbf{x} - \boldsymbol{\mu}_2) \end{aligned}$$

is quadratic function of $\mathbf{x}$, and the corresponding discrimination rule is

$$\begin{cases} \mathbf{x} \in \pi_1, & W(\mathbf{x}) \leq 0 \\ \mathbf{x} \in \pi_2, & W(\mathbf{x}) > 0 \end{cases} \quad (5.2.10)$$

- **Example 5.2.2** In example 5.2.1, suppose π_1 and π_2 have different variances as σ_1^2 and σ_2^2

$$d(x, \pi_i) = \frac{|x - \mu_i|}{\sigma_i}, \quad i = 1, 2$$

when $\mu_1 < x < \mu_2$, discrimination function can be:

$$\begin{aligned} W(x) &= d(x, \pi_1) - d(x, \pi_2) = \frac{x - \mu_1}{\sigma_1} - \frac{\mu_2 - x}{\sigma_2} \\ &= \frac{\sigma_1 + \sigma_2}{\sigma_1 \sigma_2} \left(x - \frac{\sigma_2 \mu_1 + \sigma_1 \mu_2}{\sigma_1 + \sigma_2} \right) = \frac{\sigma_1 + \sigma_2}{\sigma_1 \sigma_2} (x - \mu^*) \end{aligned}$$

where

$$\mu^* = \frac{\sigma_2 \mu_1 + \sigma_1 \mu_2}{\sigma_1 + \sigma_2}$$

is the weighted average of μ_1 and μ_2 , usually called **Threshold Value**(**閾值点**), shown in Figure 5.2.2.

► Discrimination rule

$$\begin{cases} x \in \pi_1, & x \leq \mu^* \\ x \in \pi_2, & x > \mu^* \end{cases}$$

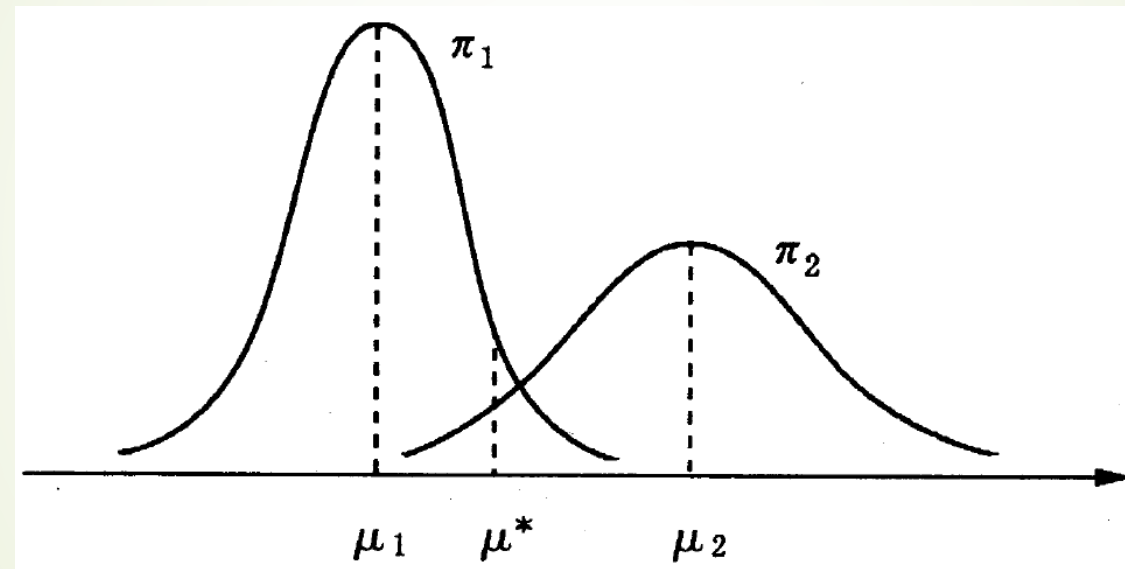


Figure 5.2.2 Threshold Value with different group variances

- In real application, μ_1, μ_2 , Σ_1 and Σ_2 are generally unknown indexes. They can be substituted by corresponding sample values.

§ 5.2.2 multiple group distance discriminant

- Suppose $\pi_1, \pi_2, \dots, \pi_k$ are k groups and their means and covariance matrices are respectively $\mu_1, \mu_2, \dots, \mu_k$ and $\Sigma_1(>0), \Sigma_2(>0), \dots, \Sigma_k(>0)$. Then the square Mahalanobis distance between population π_i and $\mathbf{x}$ is

$$d^2(\mathbf{x}, \pi_i) = (\mathbf{x} - \mu_i)' \Sigma_i^{-1} (\mathbf{x} - \mu_i), \quad i = 1, 2, \dots, k \quad (5.2.11)$$

discrimination rule: $\mathbf{x} \in \pi_l$, if $d^2(\mathbf{x}, \pi_l) = \min_{1 \leq i \leq k} d^2(\mathbf{x}, \pi_i)$ (5.2.12)

- If $\Sigma_1 = \Sigma_2 = \dots = \Sigma_k = \Sigma$, then the above discrimination rule can be furtherly simplified as

$$\begin{aligned} d^2(\mathbf{x}, \pi_i) &= (\mathbf{x} - \mu_i)' \Sigma^{-1} (\mathbf{x} - \mu_i) = \mathbf{x}' \Sigma^{-1} \mathbf{x} - 2\mu_i' \Sigma^{-1} \mathbf{x} + \mu_i' \Sigma^{-1} \mu_i \\ &= \mathbf{x}' \Sigma^{-1} \mathbf{x} - 2(I_i' \mathbf{x} + c_i), \end{aligned}$$

where $I_i = \Sigma^{-1} \mu_i$, $c_i = -\frac{1}{2} \mu_i' \Sigma^{-1} \mu_i$, $i = 1, 2, \dots, k$.

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- Discrimination rule can be furtherly simplified as:

$$\mathbf{x} \in \pi_l, \quad \text{if } \mathbf{I}'_l \mathbf{x} + c_l = \max_{1 \leq i \leq k} (\mathbf{I}'_i \mathbf{x} + c_i) \quad (5.2.13)$$

where $\mathbf{I}'_i \mathbf{x} + c_i$ is linear discriminant function.

Exercise2. When $k=2$, rewrite the above equation as: (5.2.14)

$$\begin{cases} \mathbf{x} \in \pi_1, & \text{if } \mathbf{I}'_1 \mathbf{x} + c_1 \geq \mathbf{I}'_2 \mathbf{x} + c_2 \\ \mathbf{x} \in \pi_2, & \text{if } \mathbf{I}'_1 \mathbf{x} + c_1 < \mathbf{I}'_2 \mathbf{x} + c_2 \end{cases}$$

it is equivalent to **Linear Discriminant Function rule** (5.2.3) .

Exercise2: When $k=2$, rewrite the above equation as:

$$\begin{cases} \mathbf{x} \in \pi_1, & \text{if } \mathbf{I}'_1 \mathbf{x} + c_1 \geq \mathbf{I}'_2 \mathbf{x} + c_2 \\ \mathbf{x} \in \pi_2, & \text{if } \mathbf{I}'_1 \mathbf{x} + c_1 < \mathbf{I}'_2 \mathbf{x} + c_2 \end{cases} \quad (5.2.14)$$

Proof: It is easy to obtain that

$$\begin{aligned} & \mathbf{I}'_1 \mathbf{x} + c_1 - (\mathbf{I}'_2 \mathbf{x} + c_2) \\ &= (\Sigma^{-1} \mu_1)' \mathbf{x} - 0.5 \mu'_1 \Sigma^{-1} \mu_1 - (\Sigma^{-1} \mu_2)' \mathbf{x} + 0.5 \mu'_2 \Sigma^{-1} \mu_2 \\ &= \mathbf{x}' \Sigma^{-1} (\mu_1 - \mu_2)' - 0.5 (\mu'_1 \Sigma^{-1} \mu_1 - \mu'_2 \Sigma^{-1} \mu_2) \\ &= \mathbf{x}' \Sigma^{-1} (\mu_1 - \mu_2)' - 0.5 (\mu_1 - \mu_2)' \Sigma^{-1} (\mu_1 - \mu_2) \\ &= \left(\mathbf{x} - \frac{\mu_1 + \mu_2}{2} \right)' \Sigma^{-1} (\mu_1 - \mu_2). \end{aligned}$$

Therefore,

$$\mathbf{I}'_1 \mathbf{x} + c_1 - (\mathbf{I}'_2 \mathbf{x} + c_2) = \mathbf{a}' (\mathbf{x} - \bar{\mu}) \equiv w(\mathbf{x}) \quad (5.2.3)$$

with $\mathbf{a} = \Sigma^{-1} (\mu_1 - \mu_2)$ and $\bar{\mu} = \frac{\mu_1 + \mu_2}{2}$.

In practice, $\mu_1, \mu_2, \dots, \mu_k$ and $\Sigma_1, \Sigma_2, \dots, \Sigma_k$ are usually unknown and their values can be replaced by corresponding sample estimates. Suppose $\mathbf{x}_{i1}, \mathbf{x}_{i2}, \dots, \mathbf{x}_{in_i}$ are randomly chosen from group π_i , then μ_i can be estimated as:

$$\bar{\mathbf{x}}_i = \frac{1}{n_i} \sum_{j=1}^{n_i} \mathbf{x}_{ij}$$

for $i=1, 2, \dots, k$. There exist two kinds of estimations of $\Sigma_1, \Sigma_2, \dots, \Sigma_k$.

- I. $\Sigma_1 = \Sigma_2 = \dots = \Sigma_k$. Joint unbiased estimate of Σ :

$$\mathbf{S}_p = \frac{1}{n-k} \sum_{i=1}^k (n_i - 1) \mathbf{S}_i$$

where $n = n_1 + n_2 + \dots + n_k$, $\mathbf{S}_i = \frac{1}{n_i - 1} \sum_{j=1}^{n_i} (\mathbf{x}_{ij} - \bar{\mathbf{x}}_i)(\mathbf{x}_{ij} - \bar{\mathbf{x}}_i)'$ is the sample covariance matrix of group i .

- Discrimination rule in real application is

$$\mathbf{x} \in \pi_l, \quad \text{if } \hat{\mathbf{I}}_l' \mathbf{x} + \hat{c}_l = \max_{1 \leq i \leq k} (\hat{\mathbf{I}}_i' \mathbf{x} + \hat{c}_i) \quad (5.2.15)$$

where $\hat{\mathbf{I}}_i = \mathbf{S}_p^{-1} \bar{\mathbf{x}}_i$, $\hat{c}_i = -\frac{1}{2} \bar{\mathbf{x}}_i' \mathbf{S}_p^{-1} \bar{\mathbf{x}}_i$, $i = 1, 2, \dots, k$.

II. $\Sigma_1, \Sigma_2, \dots, \Sigma_k$ not completely equal

- Σ_i can be estimated as S_i ($i=1,2,\dots,k$) with $S_i = \frac{1}{n_i - 1} \sum_{j=1}^{n_i} (\mathbf{x}_{ij} - \bar{\mathbf{x}}_i)(\mathbf{x}_{ij} - \bar{\mathbf{x}}_i)'$
- Discrimination rule in real application is

$$\mathbf{x} \in \pi_l, \quad \text{if } \hat{d}^2(\mathbf{x}, \pi_l) = \min_{1 \leq i \leq k} \hat{d}^2(\mathbf{x}, \pi_i) \quad (5.2.16)$$

where $\hat{d}^2(\mathbf{x}, \pi_i) = (\mathbf{x} - \bar{\mathbf{x}}_i)' S_i^{-1} (\mathbf{x} - \bar{\mathbf{x}}_i), \quad i = 1, 2, \dots, k$

Whether $\Sigma_1, \Sigma_2, \dots, \Sigma_k$ are assumed equal.

- In real application, $\Sigma_1, \Sigma_2, \dots, \Sigma_k$ are unlikely completely equal. What we need to concern about is whether $\Sigma_1, \Sigma_2, \dots, \Sigma_k$ are significantly different. If not, then we can assume $\Sigma_1 = \Sigma_2 = \dots = \Sigma_k = \Sigma$. Therefore, we can utilize corresponding discrimination rule. In this situation, discriminant function is linear.
- If there is doubt about $\Sigma_1 = \Sigma_2 = \dots = \Sigma_k = \Sigma$, one common method is to consider the following hypothesis test
$$H_0: \Sigma_1 = \Sigma_2 = \dots = \Sigma_k, \quad H_1: \Sigma_i \neq \Sigma_j, \text{ at least one couple } i \neq j \quad (4.7.1)$$
based on the assumption that the population of each group obeys normal distribution. **Box-M test** can be used to judge the above hypothesis test.
- If we're not sure whether we should accept H_0 , we can go through the discrimination procedure in both equal and unequal situations respectively. Then use cross validation to compare the misclassification probabilities in order to choose which one to use.

Example 5.2.3

- Collect annual financial data of bankrupt enterprises for the two years before bankruptcy, and also collect data for enterprises with good financial performance for the same period. Data involve 4 variables : x_1 =cash flow/total debt, x_2 =net revenue/total assets, x_3 =current assets(流动资产)/floating debt, x_4 =current assets/net sales. Data are listed in Table 5.2.1, $\mathbf{I}$: bankrupt enterprises, $\mathbf{II}$: non-bankrupt enterprises.

Now given one observed value of a certain enterprise with the above variables $x = (-0.16, -0.10, 1.45, 0.51)'$, please try to determine whether this group belongs to bankrupt or non-bankrupt enterprises

Table 5.2.1

Bankrupt Status Data

Number	Group	x_1	x_2	x_3	x_4	Number	Group	x_1	x_2	x_3	x_4
1	I	-0.45	-0.41	1.09	0.45	24	II	0.38	0.11	3.27	0.35
2	I	-0.56	-0.31	1.51	0.16	25	II	0.19	0.05	2.25	0.33
3	I	0.06	0.02	1.01	0.4	26	II	0.32	0.07	4.24	0.63
4	I	-0.07	-0.09	1.45	0.26	27	II	0.31	0.05	4.45	0.69
5	I	-0.1	-0.09	1.56	0.67	28	II	0.12	0.05	2.52	0.69
6	I	-0.14	-0.07	0.71	0.28	29	II	-0.02	0.02	2.05	0.35
7	I	0.04	0.01	1.5	0.71	30	II	0.22	0.08	2.35	0.4
8	I	-0.07	-0.06	1.37	0.4	31	II	0.17	0.07	1.8	0.52
9	I	0.07	-0.01	1.37	0.34	32	II	0.15	0.05	2.17	0.55
10	I	-0.14	-0.14	1.42	0.43	33	II	-0.1	-0.01	2.5	0.58
11	I	-0.23	-0.3	0.33	0.18	34	II	0.14	-0.03	0.46	0.26
12	I	0.07	0.02	1.31	0.25	35	II	0.14	0.07	2.61	0.52
13	I	0.01	0	2.15	0.7	36	II	0.15	0.06	2.23	0.56
14	I	-0.28	-0.23	1.19	0.66	37	II	0.16	0.05	2.31	0.2
15	I	0.15	0.05	1.88	0.27	38	II	0.29	0.06	1.84	0.38
16	I	0.37	0.11	1.99	0.38	39	II	0.54	0.11	2.33	0.48
17	I	-0.08	-0.08	1.51	0.42	40	II	-0.33	-0.09	3.01	0.47
18	I	0.05	0.03	1.68	0.95	41	II	0.48	0.09	1.24	0.18
19	I	0.01	0	1.26	0.6	42	II	0.56	0.11	4.29	0.44
20	I	0.12	0.11	1.14	0.17	43	II	0.2	0.08	1.99	0.3
21	I	-0.28	-0.27	1.27	0.51	44	II	0.47	0.14	2.92	0.45
22	II	0.51	0.1	2.49	0.54	45	II	0.17	0.04	2.45	0.14
23	II	0.08	0.02	2.01	0.53	46	II	0.58	0.04	5.06	0.13

➡ Use (5.2.15) to discriminate it when $\Sigma_1 = \Sigma_2$:

Discrimination rule in real application is

$$\mathbf{x} \in \pi_l, \quad \text{if } \hat{\mathbf{I}}'_l \mathbf{x} + \hat{c}_l = \max_{1 \leq i \leq k} (\hat{\mathbf{I}}'_i \mathbf{x} + \hat{c}_i) \quad (5.2.15)$$

where $\hat{\mathbf{I}}_i = \mathbf{S}_p^{-1} \bar{\mathbf{x}}_i$, $\hat{c}_i = -\frac{1}{2} \bar{\mathbf{x}}'_i \mathbf{S}_p^{-1} \bar{\mathbf{x}}_i$, $i = 1, 2, \dots, k$.

$$\bar{\mathbf{x}}_1 = \begin{pmatrix} -0.0690 \\ -0.0814 \\ 1.3667 \\ 0.4376 \end{pmatrix}, \quad \bar{\mathbf{x}}_2 = \begin{pmatrix} 0.2352 \\ 0.0556 \\ 2.5936 \\ 0.4268 \end{pmatrix}$$

$$20\mathbf{S}_1 = \begin{pmatrix} 0.8826 & 0.5695 & 0.6899 & 0.0829 \\ 0.5695 & 0.4201 & 0.5204 & 0.0688 \\ 0.6899 & 0.5204 & 3.2861 & 0.6556 \\ 0.0829 & 0.0688 & 0.6556 & 0.8916 \end{pmatrix}$$

$$24\mathbf{S}_2 = \begin{pmatrix} 1.1292 & 0.2042 & 1.7983 & -0.1609 \\ 0.2042 & 0.0570 & 0.2060 & 0.0044 \\ 1.7983 & 0.2060 & 25.1226 & 0.7832 \\ -0.1609 & 0.0044 & 0.7832 & 0.6331 \end{pmatrix}$$

Joint estimate of Σ :

$$\mathbf{S}_p = \frac{1}{44}(20\mathbf{S}_1 + 24\mathbf{S}_2) = \begin{pmatrix} 0.0457 & 0.0176 & 0.0566 & -0.0018 \\ 0.0176 & 0.0108 & 0.0165 & 0.0017 \\ 0.0566 & 0.0165 & 0.6457 & 0.0327 \\ -0.0018 & 0.0017 & 0.0327 & 0.0347 \end{pmatrix}$$

$$\mathbf{S}_p^{-1} = \begin{pmatrix} 67.9692 & -106.2364 & -3.8556 & 12.2182 \\ -106.2364 & 262.2058 & 3.6899 & -21.5137 \\ -3.8556 & 3.6899 & 1.9020 & -2.1693 \\ 12.2182 & -21.5137 & -2.1693 & 32.5632 \end{pmatrix}$$

$$\hat{\mathbf{I}}_1 = \mathbf{S}_p^{-1} \bar{\mathbf{x}}_1 = \begin{pmatrix} 4.035 \\ -18.387 \\ 1.616 \\ 12.194 \end{pmatrix}, \quad \hat{\mathbf{I}}_2 = \mathbf{S}_p^{-1} \bar{\mathbf{x}}_2 = \begin{pmatrix} 5.295 \\ -10.020 \\ 3.306 \\ 9.949 \end{pmatrix}$$

$$\hat{c}_1 = -\frac{1}{2} \bar{\mathbf{x}}_1' \mathbf{S}_p^{-1} \bar{\mathbf{x}}_1 = -4.382, \quad \hat{c}_2 = -\frac{1}{2} \bar{\mathbf{x}}_2' \mathbf{S}_p^{-1} \bar{\mathbf{x}}_2 = -6.754$$

$$\mathbf{x} = (x_1, x_2, x_3, x_4)'$$

Therefore,

$$\hat{\mathbf{I}}_1' \mathbf{x} + \hat{c}_1 = 4.035x_1 - 18.387x_2 + 1.616x_3 + 12.194x_4 - 4.382$$

$$\hat{\mathbf{I}}_2' \mathbf{x} + \hat{c}_2 = 5.295x_1 - 10.020x_2 + 3.306x_3 + 9.949x_4 - 6.754$$

For the unclassified enterprise $\mathbf{x} = (-0.16, -0.10, 1.45, 0.51)'$, we can calculate as:

$$\hat{\mathbf{I}}_1' \mathbf{x} + \hat{c}_1 = 5.373, \quad \hat{\mathbf{I}}_2' \mathbf{x} + \hat{c}_2 = 3.268$$

π_1 : bankrupt enterprises

$$\mathbf{x} \in \pi_l, \quad \text{if } \mathbf{I}_l' \mathbf{x} + c_l = \max_{1 \leq i \leq k} (\mathbf{I}_i' \mathbf{x} + c_i)$$

Based on (5.2.15), the enterprise is classified as bankrupt enterprise.

Table 5.2.3

discrimination through (5.2.7) reusing sample

discriminations		I	II
facts	I	18	3
	II	1	24

- In Table 5.2.3, misclassification probability:

$$\hat{P}(2|1) = \frac{n(2|1)}{n_1} = \frac{3}{21} = 0.143, \quad \hat{P}(1|2) = \frac{n(1|2)}{n_2} = \frac{1}{25} = 0.04$$

- Use cross validation through (5.2.8) and the discriminations are listed in Table 5.2.4.

Table 5.2.4 discrimination through (5.2.8)

discriminations facts	I	II
I	18	3
II	2	23

- In Table 5.2.4, misclassification probability:

$$\hat{P}(2|1) = \frac{3}{21} = 0.143, \quad \hat{P}(1|2) = \frac{2}{25} = 0.08$$

$$\mathbf{x} \in \pi_l, \quad \text{if } \hat{d}^2(\mathbf{x}, \pi_l) = \min_{1 \leq i \leq k} \hat{d}^2(\mathbf{x}, \pi_i)$$

- If (5.2.16) is used to discriminate when $\Sigma_1 \neq \Sigma_2$, then misclassification probability estimated based on (5.2.7) (normal group assumption) is

$$\hat{P}(2|1) = \frac{2}{21} = 0.095, \quad \hat{P}(1|2) = \frac{1}{25} = 0.04$$

- Misclassification probability estimated based on (5.2.8)(cross validation) is

$$\hat{P}(2|1) = \frac{4}{21} = 0.190, \quad \hat{P}(1|2) = \frac{1}{25} = 0.04$$

- **Note** In example 5.2.3, under the presupposition of stable economic environment during past and further two years, we can substitute the present value $\mathbf{x}=(x_1, x_2, x_3, x_4)'$ into the constructed discriminant function in this example in order to discriminate whether the enterprise will be bankrupted in two years.



Assignment: will be issued on this Thursday.





§ 5.3 Bayesian Discriminant

- § 5.3.1 maximum posterior probability
- § 5.3.2 Minimum Expected Cost of Misclassification

One counterexample of distance discriminant

➤ π_1 (postgraduate group) : $N_1=2000, \mu_1=500$

π_2 (undergraduate group) : $N_2=8000, \mu_2=400$

There are 1000 people $x \geq 500$ in π_1 while 2000 in π_2 . We observe one student with $x=500$. Now please try to discriminate the group of the student.

➤ Distance discriminant is obviously improper. Consider the following prior probability :

$$p_1 = \frac{2000}{10000} = 0.2, \quad p_2 = \frac{8000}{10000} = 0.8$$

Review:

- Suppose $B_1, B_2, \dots$ are a series of mutually incompatible events and

$$\bigcup_{i=1}^{\infty} B_i = \Omega \quad P(B_i) > 0, \quad i = 1, 2, \dots$$

Then for any event A , we have

$$P(A) = \sum_{i=1}^{\infty} P(B_i)P(A | B_i)$$

called **Total Probability Theorem**.

Bayes Rule

- Suppose $B_1, B_2, \dots$ are a series of mutually incompatible events and
- $$\bigcup_{i=1}^{\infty} B_i = \Omega \quad P(B_i) > 0, \quad i = 1, 2, \dots$$

Then for any event A , we have

$$P(B_i \mid A) = \frac{P(B_i)P(A \mid B_i)}{\sum_{i=1}^{\infty} P(B_i)P(A \mid B_i)}, \quad i = 1, 2, \dots$$

Posterior probability

Prior probability

§ 5.3.1 maximum posterior probability

- Suppose $\pi_1, \pi_2, \dots, \pi_k$ are k groups and the probability density function of group π_i is $f_i(\mathbf{x})$. Prior probability of sample $\mathbf{x}$ from π_i is $p_i, i=1, 2, \dots, k$, satisfying $p_1 + p_2 + \dots + p_k = 1$. Then posterior probabilities of $\mathbf{x}$ belonging to π_i are

$$P(\pi_i | \mathbf{x}) = \frac{p_i f_i(\mathbf{x})}{\sum_{i=1}^k p_i f_i(\mathbf{x})}, \quad i = 1, 2, \dots, k$$

where $p_i = P(\pi_i)$, $f_i(x) = P(X | \pi_i)$.

- Maximum posterior probability criterion's discriminant rule:

$$\mathbf{x} \in \pi_l, \quad \text{if } P(\pi_l | \mathbf{x}) = \max_{1 \leq i \leq k} P(\pi_i | \mathbf{x}) \quad (5.3.2)$$

- **Example 5.3.1** Suppose π_1 , π_2 and π_3 are 3 groups. Try to discriminate sample $\mathbf{x}_0$ and $p_1=0.05$, $p_2=0.65$, $p_3=0.30$, $f_1(\mathbf{x}_0)=0.10$, $f_2(\mathbf{x}_0)=0.63$, $f_3(\mathbf{x}_0)=2.4$ are given. Now calculate posterior probability of $\mathbf{x}_0$ belonging to each group :

$$\begin{aligned} P(\pi_1 | \mathbf{x}_0) &= \frac{p_1 f_1(\mathbf{x}_0)}{\sum_{i=1}^3 p_i f_i(\mathbf{x}_0)} = \frac{0.05 \times 0.10}{0.05 \times 0.10 + 0.65 \times 0.63 + 0.30 \times 2.4} \\ &= \frac{0.005}{1.1345} = 0.004 \end{aligned}$$

$$P(\pi_2 | \mathbf{x}_0) = \frac{p_2 f_2(\mathbf{x}_0)}{\sum_{i=1}^3 p_i f_i(\mathbf{x}_0)} = \frac{0.65 \times 0.63}{1.1345} = 0.361$$

$$P(\pi_3 | \mathbf{x}_0) = \frac{p_3 f_3(\mathbf{x}_0)}{\sum_{i=1}^3 p_i f_i(\mathbf{x}_0)} = \frac{0.30 \times 2.4}{1.1345} = 0.635$$

So, $\mathbf{x}_0$ should be discriminated to π_3 .

Normal Groups

- Suppose $\pi_i \sim N_p(\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i)$, $\boldsymbol{\Sigma}_i > 0$, $i=1,2,\dots,k$. The probability density of π_i is:

$$f_i(\mathbf{x}) = (2\pi)^{-p/2} |\boldsymbol{\Sigma}_i|^{-1/2} \exp [-0.5 d^2(\mathbf{x}, \pi_i)]$$

where

$$d^2(\mathbf{x}, \pi_i) = (\mathbf{x} - \boldsymbol{\mu}_i)' \boldsymbol{\Sigma}_i^{-1} (\mathbf{x} - \boldsymbol{\mu}_i)$$

is the square Mahalanobis distance between $\mathbf{x}$ and π_i .

- Specific calculation formulas of posterior probabilities on the following conditions:

- When $p_1=p_2=\cdots=p_k=1/k$, $\Sigma_1=\Sigma_2=\cdots=\Sigma_k=\Sigma$,

$$P(\pi_i | \mathbf{x}) = \frac{\exp\left[-\frac{1}{2}d^2(\mathbf{x}, \pi_i)\right]}{\sum_{i=1}^k \exp\left[-\frac{1}{2}d^2(\mathbf{x}, \pi_i)\right]}$$
$$P(\pi_i | \mathbf{x}) = \frac{p_i f_i(\mathbf{x})}{\sum_{i=1}^k p_i f_i(\mathbf{x})}, \quad i = 1, 2, \dots, k$$

- When $p_1=p_2=\cdots=p_k=1/k$, $\Sigma_1, \Sigma_2, \dots, \Sigma_k$ are not completely equal,

$$P(\pi_i | \mathbf{x}) = \frac{\exp\left\{-\frac{1}{2}\left[d^2(\mathbf{x}, \pi_i) + \ln|\Sigma_i|\right]\right\}}{\sum_{i=1}^k \exp\left\{-\frac{1}{2}\left[d^2(\mathbf{x}, \pi_i) + \ln|\Sigma_i|\right]\right\}}$$

- When $\Sigma_1 = \Sigma_2 = \dots = \Sigma_k = \Sigma$, $p_1, p_2, \dots, p_k$ are not completely equal,

$$P(\pi_i | \mathbf{x}) = \frac{\exp\left\{-\frac{1}{2}\left[d^2(\mathbf{x}, \pi_i) - 2\ln p_i\right]\right\}}{\sum_{i=1}^k \exp\left\{-\frac{1}{2}\left[d^2(\mathbf{x}, \pi_i) - 2\ln p_i\right]\right\}}$$

- When both $p_1, p_2, \dots, p_k$ and $\Sigma_1, \Sigma_2, \dots, \Sigma_k$ are not completely equal ,

$$P(\pi_i | \mathbf{x}) = \frac{\exp\left\{-\frac{1}{2}\left[d^2(\mathbf{x}, \pi_i) + \ln |\Sigma_i| - 2\ln p_i\right]\right\}}{\sum_{i=1}^k \exp\left\{-\frac{1}{2}\left[d^2(\mathbf{x}, \pi_i) + \ln |\Sigma_i| - 2\ln p_i\right]\right\}}$$

➤ Posterior probabilities of the above situations are shown as:

$$P(\pi_i | \mathbf{x}) = \frac{\exp\left[-\frac{1}{2}D^2(\mathbf{x}, \pi_i)\right]}{\sum_{i=1}^k \exp\left[-\frac{1}{2}D^2(\mathbf{x}, \pi_i)\right]}, \quad i = 1, 2, \dots, k \quad (5.3.4)$$

$$\text{where, } D^2(x, \pi_i) = d^2(x, \pi_i) + g_i + h_i \quad (5.3.5)$$

$$g_i = \begin{cases} \ln|\Sigma_i|, & \text{if } \Sigma_1, \Sigma_2, \dots, \Sigma_k \text{ are not completely equal} \\ 0, & \text{if } \Sigma_1 = \Sigma_2 = \dots = \Sigma_k = \Sigma \end{cases}$$

$$h_i = \begin{cases} -2\ln p_i, & \text{if } p_1, p_2, \dots, p_k \text{ are not completely equal} \\ 0, & \text{if } p_1 = p_2 = \dots = p_k = \frac{1}{k} \end{cases}$$

$$i = 1, 2, \dots, k$$

- $D^2(\mathbf{x}, \pi_i)$ is called **Generalized Square Distance** between $\mathbf{x}$ and π_i . Under the normal hypothesis, the above discriminant rule can be equally shown as:

$$\mathbf{x} \in \pi_l, \quad \text{if } D^2(\mathbf{x}, \pi_l) = \min_{1 \leq i \leq k} D^2(\mathbf{x}, \pi_i) \quad (5.3.6)$$

from (5.3.4).

- When $\Sigma_1 = \Sigma_2 = \dots = \Sigma_k = \Sigma$, the above posterior probability formula (5.3.4) can be simplified as:

$$P(\pi_i | \mathbf{x}) = \frac{\exp(\mathbf{I}'_i \mathbf{x} + c_i + \ln p_i)}{\sum_{i=1}^k \exp(\mathbf{I}'_i \mathbf{x} + c_i + \ln p_i)}, \quad i = 1, 2, \dots, k \quad (5.3.7)$$

➤ $-0.5d^2(\mathbf{x}, \pi_i) = -0.5\mathbf{x}'\Sigma^{-1}\mathbf{x} + (\mathbf{I}'_i \mathbf{x} + c_i).$

where $\mathbf{I}_i = \Sigma^{-1}\boldsymbol{\mu}_i$, $c_i = -0.5\boldsymbol{\mu}'_i \Sigma^{-1}\boldsymbol{\mu}_i$, $i=1, 2, \dots, k$. On this situation, Discriminant rule is equivalent to

$$\mathbf{x} \in \pi_l, \quad \text{if } \mathbf{I}'_l \mathbf{x} + c_l + \ln(p_l) = \max_{1 \leq i \leq k} (\mathbf{I}'_i \mathbf{x} + c_i + \ln(p_i)) \quad (5.3.8)$$

- If we know nothing about the prior information from which $\mathbf{x}$ belongs to, the same priors $p_1=p_2=\dots=p_k=1/k$ can be taken. **Discriminant rule can be simplified as multiple group distance discriminant.**

$$\mathbf{x} \in \pi_l, \quad \text{if } \mathbf{I}'_l \mathbf{x} + c_l = \max_{1 \leq i \leq k} (\mathbf{I}'_i \mathbf{x} + c_i) \quad (5.2.13)$$

where $\mathbf{I}_i = \Sigma^{-1} \boldsymbol{\mu}_i$, $c_i = -0.5 \boldsymbol{\mu}_i' \Sigma^{-1} \boldsymbol{\mu}_i$, $i=1,2,\dots,k$.

- In real application, $\boldsymbol{\mu}_i$ and Σ_i ($i=1,2,\dots,k$) of the equations (5.3.6) are unknown and need to be replaced by corresponding sample estimates.◦

► **Example 5.3.2** In example 5.2.3, known bankrupt enterprises is approximately 10%, i.e., $p_1=0.1$, $p_2=0.9$. Assume they are both normal groups and $\Sigma_1=\Sigma_2=\Sigma$. Please find the posterior probability of indiscriminant enterprise $\mathbf{x}=(-0.16, -0.10, 1.45, 0.51)'$. $\hat{\mathbf{I}}_1'\mathbf{x} + \hat{c}_1 = 5.373$, $\hat{\mathbf{I}}_2'\mathbf{x} + \hat{c}_2 = 3.268$
 Solution: From equation (5.3.6), we have $\hat{\mathbf{I}}_i = \mathbf{S}_p^{-1}\bar{\mathbf{x}}_i, \hat{c}_i = -\frac{1}{2}\bar{\mathbf{x}}_i'\mathbf{S}_p^{-1}\bar{\mathbf{x}}_i, i = 1, 2, \dots, k$

$$P(\pi_1 | \mathbf{x}) = \frac{\exp(\hat{\mathbf{I}}_1'\mathbf{x} + \hat{c}_1 + \ln p_1)}{\exp(\hat{\mathbf{I}}_1'\mathbf{x} + \hat{c}_1 + \ln p_1) + \exp(\hat{\mathbf{I}}_2'\mathbf{x} + \hat{c}_2 + \ln p_2)}$$

$$= \frac{\exp(5.373 + \ln 0.1)}{\exp(5.373 + \ln 0.1) + \exp(3.268 + \ln 0.9)} = \frac{e^{3.07}}{e^{3.07} + e^{3.163}} = \frac{21.542}{45.183} = 0.477$$

$$P(\pi_2 | \mathbf{x}) = \frac{\exp(\hat{\mathbf{I}}_2'\mathbf{x} + \hat{c}_2 + \ln p_2)}{\exp(\hat{\mathbf{I}}_1'\mathbf{x} + \hat{c}_1 + \ln p_1) + \exp(\hat{\mathbf{I}}_2'\mathbf{x} + \hat{c}_2 + \ln p_2)} = \frac{23.641}{45.183} = 0.523$$

Since $P(\pi_1|\mathbf{x}) < P(\pi_2|\mathbf{x})$, this enterprise is discriminated as non-bankrupt enterprise, which is the opposite of example 5.2.3 result. This is exactly the consequence of prior probability.

小结：贝叶斯原理（最大后验概率准则）在推理和分类中的应用

- 分类是人工智能领域常见的任务之一，贝叶斯原理（最大概率准则）在分类中有广泛的应用。以下是贝叶斯原理在分类中的应用案例：
- 1. 垃圾邮件（过滤）分类：贝叶斯原理可以根据邮件中的关键词和先验知识来计算某封邮件是垃圾邮件的后验概率，如果该后验概率高于某个阈值，则可以将其分类为垃圾邮件。
- 2. 语音识别：在语音识别中，贝叶斯原理可以用来计算某个词或短语出现在语音信号中的概率。通过先验知识和语音信号的特征参数，就可以利用贝叶斯原理来对语音进行识别
- 3. 图像分类：在图像识别中，贝叶斯原理可以根据图像特征和先验知识来计算某个图像属于哪个类别的概率。通过比较概率大小，可以对图像进行分类。
- 4. 文本分类：贝叶斯原理可以应用于文本分类任务中。通过计算文本中的词语出现的概率和先验知识，可以对文本进行分类，例如将新闻文本分类为体育、政治、娱乐等类别。
- 5. 情感分析：贝叶斯原理可以应用于情感分析任务中。通过计算文本中词语的情感倾向性和先验知识，可以对文本进行情感正负分类，如判断一条评论是正面还是负面。



Assignment: See the assignment file in BB



§ 5.3.2 Minimum Expected Cost of Misclassification

➤ Example :

π_1 : qualified medicine, π_2 : unqualified medicine

For new sample $\mathbf{x}$

$$P(\pi_1 | \mathbf{x}) = 0.6, \quad P(\pi_2 | \mathbf{x}) = 0.4$$

The lost caused by two kinds of misclassification are significantly different. So, it is not proper to discriminate only based on posterior probabilities.

- 1.two groups
- 2.multiple groups

1. Two Groups

- Suppose probability density functions of π_1 and π_2 are respectively $f_1(\mathbf{x})$ and $f_2(\mathbf{x})$ and prior probabilities of π_1 and π_2 are separately p_1 and p_2 , $p_1+p_2=1$. Suppose the cost of misclassifying $\mathbf{x}$ from π_i to π_l is $c(l|i)$, $l,i=1,2$, cost matrix is

		判别为	
		π_1	π_2
真实组	π_1	0	$c(2 1)$
	π_2	$c(1 2)$	0

- For a given discriminant rule, suppose

$$R_1 = \{\mathbf{x} : \mathbf{x} \in \pi_1\}, \quad R_2 = \{\mathbf{x} : \mathbf{x} \in \pi_2\}$$

Obviously,

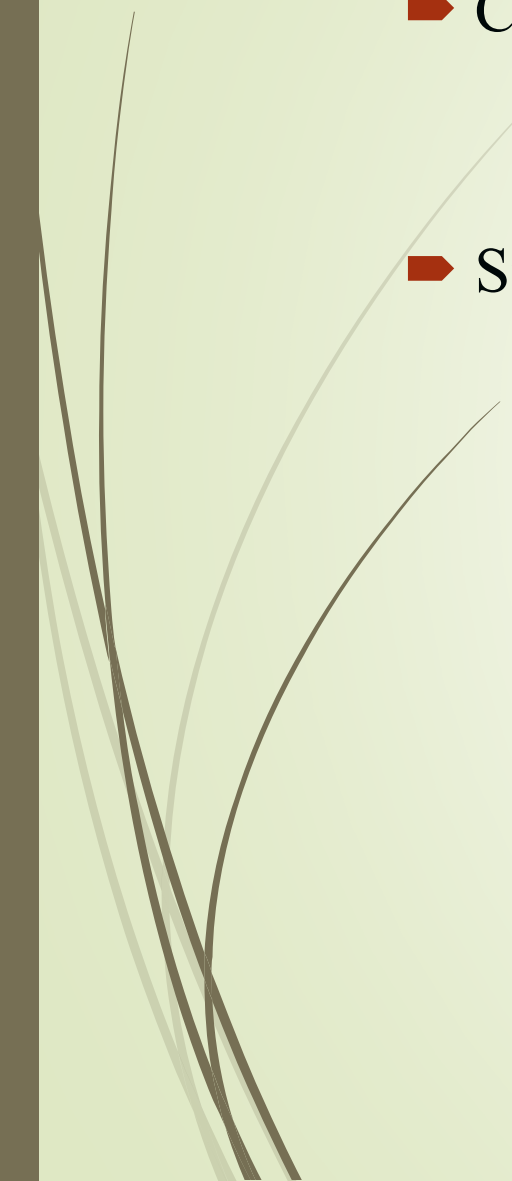
$$R_1 \cap R_2 = \Phi, \quad R_1 \cup R_2 = \Omega$$

$$\mathbf{x} \in R_1 \Leftrightarrow \mathbf{x} \in \pi_1, \quad \mathbf{x} \in R_2 \Leftrightarrow \mathbf{x} \in \pi_2$$

- 
- Conditional probability of misclassifying sample $\mathbf{x}$ of π_1 to π_2

$$P(2|1) = P(\mathbf{x} \in R_2 | \mathbf{x} \in \pi_1) = \int_{R_2} f_1(\mathbf{x}) d\mathbf{x}$$

- Similarly, conditional probability of misclassifying sample $\mathbf{x}$ of π_2 to π_1

$$P(1|2) = P(\mathbf{x} \in R_1 | \mathbf{x} \in \pi_2) = \int_{R_1} f_2(\mathbf{x}) d\mathbf{x}$$


- **Expected Cost of Misclassification** (平均误判代价), denoted as *ECM*. To memory

$$\begin{aligned} ECM &= E[c(l|i)] = c(1|1)P(i=1, l=1) + c(2|1)P(i=1, l=2) \\ &\quad + c(1|2)P(i=2, l=1) + c(2|2)P(i=2, l=2) \\ &= c(2|1)P(\mathbf{x} \in \pi_1, \mathbf{x} \in R_2) + c(1|2)P(\mathbf{x} \in \pi_2, \mathbf{x} \in R_1) \\ &= c(2|1)P(\mathbf{x} \in R_2 | \mathbf{x} \in \pi_1)P(\mathbf{x} \in \pi_1) + c(1|2)P(\mathbf{x} \in R_1 | \mathbf{x} \in \pi_2)P(\mathbf{x} \in \pi_2) \\ &= \underbrace{c(2|1)P(2|1)}_{\text{Cost 1}} p_1 + \underbrace{c(1|2)P(1|2)}_{\text{Cost 2}} p_2 \end{aligned} \tag{5.3.12}$$

Average loss of misclassification weighted by prior probability
(按先验概率加权的平均误判损失)

- **Expected Cost of Misclassification** (平均误判代价), denoted as *ECM*. To memory

$$ECM = c(2|1)P(2|1)p_1 + c(1|2)P(1|2)p_2 \quad (5.3.12)$$

Average loss of misclassification weighted by prior probability

- Minimum Expected Cost of Misclassification is the discriminant rule that minimizes *ECM*, which is given by

$$\begin{cases} \mathbf{x} \in \pi_1, & \text{if } \frac{f_1(\mathbf{x})}{f_2(\mathbf{x})} \geq \frac{c(1|2)p_2}{c(2|1)p_1} \\ \mathbf{x} \in \pi_2, & \text{if } \frac{f_1(\mathbf{x})}{f_2(\mathbf{x})} < \frac{c(1|2)p_2}{c(2|1)p_1} \end{cases} \quad (5.3.13)$$

Proof: To illustrate intuitively, we prove (5.3.13) as follows:

$$\begin{aligned} ECM &= p_1 c(2|1) \int_{R_2} f_1(x) dx + p_2 c(1|2) \int_{R_1} f_2(x) dx \\ &= p_1 c(2|1) \int_{R-R_1} f_1(x) dx + p_2 c(1|2) \int_{R_1} f_2(x) dx \\ &= p_1 c(2|1) - p_1 c(2|1) \int_{R_1} f_1(x) dx + p_2 c(1|2) \int_{R_1} f_2(x) dx \\ &= p_1 c(2|1) + \int_{R_1} (p_2 c(1|2) f_2(x) - p_1 c(2|1) f_1(x)) dx. \end{aligned}$$

From this, it can be seen that in order to minimize ECM, the integrand function needs to be negative on R_1 . This leads to the following division

$$\begin{aligned} R_1 &= \{x | p_2 c(1|2) f_2(x) - p_1 c(2|1) f_1(x) \leq 0\} \text{ and} \\ R_2 &= \{x | p_2 c(1|2) f_2(x) - p_1 c(2|1) f_1(x) > 0\}. \end{aligned}$$

Therefore, the discriminant rule of ECM is

$$\begin{cases} \mathbf{x} \in \pi_1, & \text{if } \frac{f_1(\mathbf{x})}{f_2(\mathbf{x})} \geq \frac{c(1|2)p_2}{c(2|1)p_1} \\ \mathbf{x} \in \pi_2, & \text{if } \frac{f_1(\mathbf{x})}{f_2(\mathbf{x})} < \frac{c(1|2)p_2}{c(2|1)p_1} \end{cases} \quad (5.3.13)$$

Misclassification Cost Ratio

- Minimum *ECM* needs 3 ratios : density function ratio, misclassification cost ratio and prior probability ratio. In these ratios, misclassification cost ratio has practical significance. Because in many applications, there is difficulty in determining misclassification cost directly but it is relatively easy to define misclassification cost ratio.

- Example 1

π_1 : should perform an operation,

π_2 : shouldn't perform an operation

- Example 2

π_1 : Apply for Ph.D after graduating with a bachelor's degree

π_2 : Apply for Master after graduating with a bachelor's degree

π_3 : Hunt a job after graduating with a bachelor's degree

Special Situations for (5.3.13)

➤ (1) When $p_1=p_2=0.5$, (5.3.13) can be simplified as

$$\begin{cases} \mathbf{x} \in \pi_1, & \text{if } \frac{f_1(\mathbf{x})}{f_2(\mathbf{x})} \geq \frac{c(1|2)}{c(2|1)} \\ \mathbf{x} \in \pi_2, & \text{if } \frac{f_1(\mathbf{x})}{f_2(\mathbf{x})} < \frac{c(1|2)}{c(2|1)} \end{cases} \quad \begin{cases} \mathbf{x} \in \pi_1, & \text{if } \frac{f_1(\mathbf{x})}{f_2(\mathbf{x})} \geq \frac{c(1|2)p_2}{c(2|1)p_1} \\ \mathbf{x} \in \pi_2, & \text{if } \frac{f_1(\mathbf{x})}{f_2(\mathbf{x})} < \frac{c(1|2)p_2}{c(2|1)p_1} \end{cases}$$

(5.3.13)

➤ In real application, if prior probabilities are unknown, then they are usually taken equal.

- (2) When $c(1|2) = c(2|1)$, (5.3.13) can be simplified as

$$\begin{cases} \mathbf{x} \in \pi_1, & \text{if } p_1 f_1(\mathbf{x}) \geq p_2 f_2(\mathbf{x}) \\ \mathbf{x} \in \pi_2, & \text{if } p_1 f_1(\mathbf{x}) < p_2 f_2(\mathbf{x}) \end{cases} \longrightarrow \begin{cases} \mathbf{x} \in \pi_1, & \text{if } P(\pi_1|\mathbf{x}) \geq P(\pi_2|\mathbf{x}) \\ \mathbf{x} \in \pi_2, & \text{if } P(\pi_1|\mathbf{x}) < P(\pi_2|\mathbf{x}) \end{cases}$$

- This equation is equivalent to Maximum posterior probability criterion's discriminant rule (5.3.2) with $k = 2$.

$$P(\pi_i|\mathbf{x}) = \frac{p_i f_i(\mathbf{x})}{\sum_{i=1}^k p_i f_i(\mathbf{x})}, \quad i = 1, 2, \dots, k$$

- In practice, if misclassification cost ratio cannot be confirmed, then usually taken as 1.

- (3) When $\frac{p_1}{p_2} = \frac{c(1|2)}{c(2|1)}$, (5.3.13) can be simplified as

$$\begin{cases} \mathbf{x} \in \pi_1, & \text{if } f_1(\mathbf{x}) \geq f_2(\mathbf{x}) \\ \mathbf{x} \in \pi_2, & \text{if } f_1(\mathbf{x}) < f_2(\mathbf{x}) \end{cases} \quad \begin{cases} \mathbf{x} \in \pi_1, & \text{if } \frac{f_1(\mathbf{x})}{f_2(\mathbf{x})} \geq \frac{c(1|2)p_2}{c(2|1)p_1} \\ \mathbf{x} \in \pi_2, & \text{if } \frac{f_1(\mathbf{x})}{f_2(\mathbf{x})} < \frac{c(1|2)p_2}{c(2|1)p_1} \end{cases} \quad (5.3.13)$$

In order to discriminate the belonging of new sample $\mathbf{x}_0$, we only need to compare two probability density values, $f_1(\mathbf{x}_0)$ and $f_2(\mathbf{x}_0)$.

- If apply discrimination rule (3) to example 5.2.2, then in figure 5.2.2, threshold point will shift to under the crosspoint of two density curves, m .

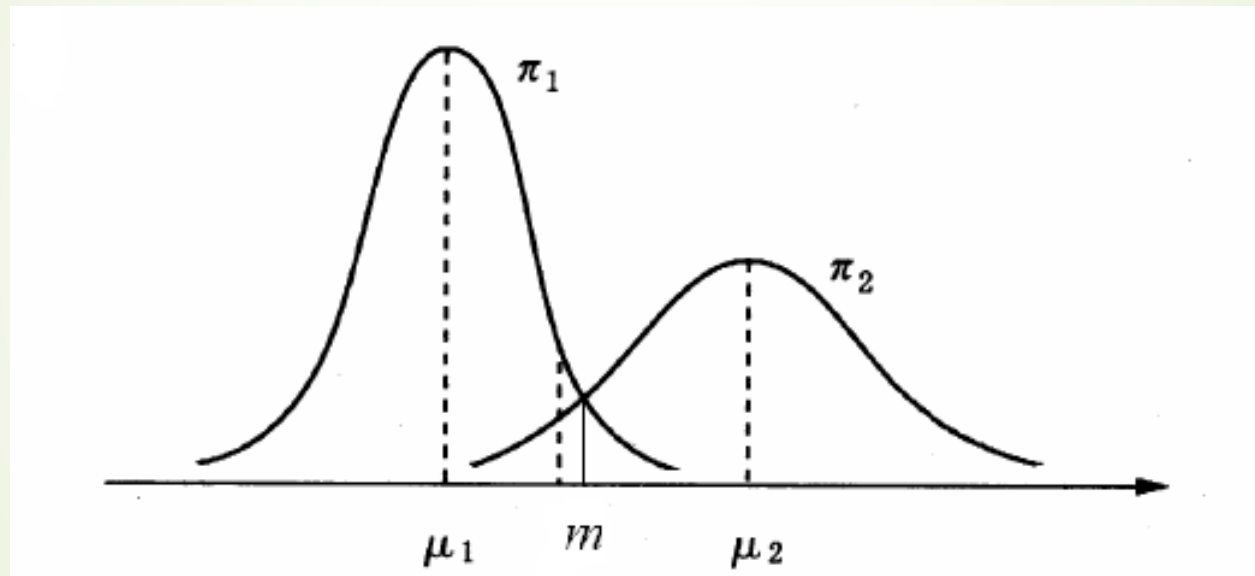


Figure 5.2.2 Threshold Value of Two Groups with Different Variances

- **Example 5.3.3** Suppose the probability density functions of π_1 and π_2 , are $f_1(\mathbf{x})$ and $f_2(\mathbf{x})$ and $c(1|2)=12$ units, $c(2|1)=4$ units. Based on the former experience, $p_1=0.6$, $p_2=0.4$, then minimum *ECM* discrimination rule

$$\left\{ \begin{array}{l} \mathbf{x} \in \pi_1, \quad \text{if } \frac{f_1(\mathbf{x})}{f_2(\mathbf{x})} \geq \frac{12 \times 0.4}{4 \times 0.6} = 2 \\ \mathbf{x} \in \pi_2, \quad \text{if } \frac{f_1(\mathbf{x})}{f_2(\mathbf{x})} < \frac{12 \times 0.4}{4 \times 0.6} = 2 \end{array} \right. \quad \left\{ \begin{array}{l} \mathbf{x} \in \pi_1, \quad \text{if } \frac{f_1(\mathbf{x})}{f_2(\mathbf{x})} \geq \frac{c(1|2)p_2}{c(2|1)p_1} \\ \mathbf{x} \in \pi_2, \quad \text{if } \frac{f_1(\mathbf{x})}{f_2(\mathbf{x})} < \frac{c(1|2)p_2}{c(2|1)p_1} \end{array} \right. \quad (5.3.13)$$

Suppose a new sample $\mathbf{x}_0$, $f_1(\mathbf{x}_0)=0.36$, $f_2(\mathbf{x}_0)=0.24$, then

$$\frac{f_1(\mathbf{x}_0)}{f_2(\mathbf{x}_0)} = \frac{0.36}{0.24} = 1.5 < 2$$

Therefore, $\mathbf{x}_0$ is discriminated to π_2 .

- In statistic application, discriminant method based on normal population plays a leading role. Now assume $\pi_i \sim N_p(\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i)$, $\boldsymbol{\Sigma}_i > 0$, $i=1,2$.

- When $\boldsymbol{\Sigma}_1 = \boldsymbol{\Sigma}_2 = \boldsymbol{\Sigma}$, the ECM rule (5.3.13) can be formulated as

$$\begin{cases} \mathbf{x} \in \pi_1, & \text{if } \mathbf{a}'(\mathbf{x} - \bar{\boldsymbol{\mu}}) \geq \ln \left[\frac{c(1|2)p_2}{c(2|1)p_1} \right] \\ \mathbf{x} \in \pi_2, & \text{if } \mathbf{a}'(\mathbf{x} - \bar{\boldsymbol{\mu}}) < \ln \left[\frac{c(1|2)p_2}{c(2|1)p_1} \right] \end{cases} \quad (5.3.18) \text{ Exercise 1}$$

$$\begin{cases} \mathbf{x} \in \pi_1, & \text{if } \frac{f_1(\mathbf{x})}{f_2(\mathbf{x})} \geq \frac{c(1|2)p_2}{c(2|1)p_1} \\ \mathbf{x} \in \pi_2, & \text{if } \frac{f_1(\mathbf{x})}{f_2(\mathbf{x})} < \frac{c(1|2)p_2}{c(2|1)p_1} \end{cases} \quad (5.3.13)$$

where $\mathbf{a} = \boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)$, $\bar{\boldsymbol{\mu}} = \frac{1}{2}(\boldsymbol{\mu}_1 + \boldsymbol{\mu}_2)$

- When $p_1 = p_2$, $c(1|2) = c(2|1)$, the above equation will be degraded as (5.2.3) .

Exercise 1 : Prove (5.3.18)

It is easy to obtain that

$$\begin{aligned} & -\frac{1}{2}(x - \mu_1)' \Sigma^{-1}(x - \mu_1) + \frac{1}{2}(x - \mu_2)' \Sigma^{-1}(x - \mu_2) \\ &= [x - \frac{1}{2}(\mu_1 + \mu_2)]' \Sigma^{-1}(\mu_1 - \mu_2) \\ &= a'(x - \bar{\mu}). \end{aligned}$$

Therefore, we have

$$\frac{\exp\{-\frac{1}{2}(x - \mu_1)' \Sigma^{-1}(x - \mu_1)\}}{\exp\{-\frac{1}{2}(x - \mu_2)' \Sigma^{-1}(x - \mu_2)\}} = \exp\{a'(x - \bar{\mu})\}.$$

By (5.3.13), we obtain

$$\begin{cases} \mathbf{x} \in \pi_1, & \text{if } a'(\mathbf{x} - \bar{\mu}) \geq \ln \left[\frac{c(1|2)p_2}{c(2|1)p_1} \right] \\ \mathbf{x} \in \pi_2, & \text{if } a'(\mathbf{x} - \bar{\mu}) < \ln \left[\frac{c(1|2)p_2}{c(2|1)p_1} \right] \end{cases} \quad (5.3.18)$$

➡ When $\Sigma_1 \neq \Sigma_2$, (5.3.13) can be written as

$$\begin{cases} \mathbf{x} \in \pi_1, & \text{if } \frac{f_1(\mathbf{x})}{f_2(\mathbf{x})} \geq \frac{c(1|2)p_2}{c(2|1)p_1} \\ \mathbf{x} \in \pi_2, & \text{if } \frac{f_1(\mathbf{x})}{f_2(\mathbf{x})} < \frac{c(1|2)p_2}{c(2|1)p_1} \end{cases} \quad (5.3.13)$$

$$\begin{cases} \mathbf{x} \in \pi_1, & \text{if } d^2(\mathbf{x}, \pi_1) - d^2(\mathbf{x}, \pi_2) \leq 2 \ln \left[\frac{c(2|1)p_1 |\Sigma_2|^{1/2}}{c(1|2)p_2 |\Sigma_1|^{1/2}} \right] \\ \mathbf{x} \in \pi_2, & \text{if } d^2(\mathbf{x}, \pi_1) - d^2(\mathbf{x}, \pi_2) > 2 \ln \left[\frac{c(2|1)p_1 |\Sigma_2|^{1/2}}{c(1|2)p_2 |\Sigma_1|^{1/2}} \right] \end{cases} \quad (5.3.18)$$

where $d^2(\mathbf{x}, \pi_i) = (\mathbf{x} - \boldsymbol{\mu}_i)' \Sigma_i^{-1} (\mathbf{x} - \boldsymbol{\mu}_i)$, $i=1,2$.

➤ When $|\Sigma_1| = |\Sigma_2|$, $p_1 = p_2$, $c(1|2) = c(2|1)$, the above equation can be degenerated as discrimination classification rule (5.2.9).

$$\begin{cases} \mathbf{x} \in \pi_1, & d^2(\mathbf{x}, \pi_1) \leq d^2(\mathbf{x}, \pi_2) \\ \mathbf{x} \in \pi_2, & d^2(\mathbf{x}, \pi_1) > d^2(\mathbf{x}, \pi_2) \end{cases} \quad (5.2.9)$$

2. ECM for Multiple Groups

➤ Suppose $f_i(\mathbf{x})$ is the probability density function of π_i , $i=1,2,\dots,k$.

➤ Notations:

p_i : prior probability of π_i , $i=1,2,\dots,k$.

$c(l|i)$: cost of misclassifying $\mathbf{x}$ of π_i to π_l , $l,i=1,2,\dots,k$, for $l=i, c(i|i)=0$

R_l : the set of $\mathbf{x}$ discriminated to π_l , $l=1,2,\dots,k$.

Thus, for l , $i=1,2,\dots,k$, the conditional probability of misclassifying $\mathbf{x}$ of π_i to π_l :

$$P(l | i) = P(\mathbf{x} \in R_l | \mathbf{x} \in \pi_i) = \int_{R_l} f_i(\mathbf{x}) d\mathbf{x}$$

2. ECM for Multiple Groups

$$\begin{aligned} ECM &= E[c(l|i)] \\ &= \sum_{i=1}^k \sum_{l=1}^k c(l|i) P(\mathbf{x} \in \pi_i, \mathbf{x} \in R_l) \\ &= \sum_{i=1}^k \sum_{l=1}^k c(l|i) P(\mathbf{x} \in R_l | \mathbf{x} \in \pi_i) P(\mathbf{x} \in \pi_i) \\ &= \sum_{i=1}^k \sum_{l=1}^k c(l|i) P(l|i) p_i = \sum_{i=1}^k p_i \sum_{\substack{l=1 \\ l \neq i}}^k c(l|i) P(l|i) \end{aligned}$$

Average loss of misclassification weighted by prior probability
(按先验概率加权的平均误判损失)

➤ Minimum *ECM* discriminant rule is

$$\mathbf{x} \in \pi_l, \quad \text{if } \sum_{\substack{j=1 \\ j \neq l}}^k p_j f_j(\mathbf{x}) c(l|j) = \min_{1 \leq i \leq k} \sum_{\substack{j=1 \\ j \neq i}}^k p_j f_j(\mathbf{x}) c(i|j) \quad (5.3.23)$$

➤ Assume all the misclassification costs are equal, without loss of generality, let $c(l|i)=1, l \neq i, l, i=1, 2, \dots, k$, then

$$ECM = \sum_{i=1}^k p_i \sum_{\substack{l=1 \\ l \neq i}}^k P(l|i) = 1 - \sum_{i=1}^k p_i P(i|i) \quad \begin{matrix} \text{Total Correct Discrimination Probability} \\ \sum_{l=1}^k P(l|i) = 1 \end{matrix} \quad (5.3.24)$$

is the sum of all the misclassification probabilities, called **Total Misclassification Probability**(总的误判概率). Thus, the minimum expected cost of misclassification criterion is called **Minimum Total Misclassification Probability Criterion**(最小总误判概率准则), and the equation (5.3.23) can be simplified as:

$$\mathbf{x} \in \pi_l, \quad \text{if } p_l f_l(\mathbf{x}) = \max_{1 \leq i \leq k} p_i f_i(\mathbf{x}) \quad p(\pi_i|\mathbf{x}) \propto p(\pi_i)p(\mathbf{x}|\pi_i) = p_i f_i(\mathbf{x}) \quad (5.3.26)$$

➤ It is equivalent to (5.3.2). Namely, **ECM criterion** is equivalent to maximum posterior probability criterion when all the misclassification costs are equal (To remember this property).

Proof of (5.3.24):

Suppose $B = \{\text{misclassification}\}$, $A_i = \{\text{samples from } \pi_i\}$, $i=1,2,\dots,k$

Use total probability theorem to obtain **Total Misclassification Probability** (总的误判概率)

$$P(B) = \sum_{i=1}^k P(A_i) P(B | A_i) = \sum_{i=1}^k p_i \sum_{\substack{l=1 \\ l \neq i}}^k P(l | i)$$

In addition, **Total Correct Discrimination Probability** (总的正确判别概率)

$$\begin{aligned} P(\bar{B}) &= 1 - P(B) = 1 - \sum_{i=1}^k p_i \sum_{\substack{l=1 \\ l \neq i}}^k P(l | i) \\ &= 1 - \sum_{i=1}^k p_i [1 - P(i | i)] = \sum_{i=1}^k p_i P(i | i) \end{aligned}$$

► **Example 5.3.4** In example 5.3.1, suppose misclassification cost matrix

		判别为		
		π_1	π_2	π_3
真实组	π_1	$c(1 1) = 0$	$c(2 1) = 10$	$c(3 1) = 200$
	π_2	$c(1 2) = 20$	$c(2 2) = 0$	$c(3 2) = 100$
	π_3	$c(1 3) = 60$	$c(2 3) = 50$	$c(3 3) = 0$

► Use minimum *ECM* criterion (5.3.23) to discriminate:

$$\begin{aligned} l=1: & p_2 f_2(\mathbf{x}_0) c(1|2) + p_3 f_3(\mathbf{x}_0) c(1|3) \\ &= 0.65 \times 0.63 \times 20 + 0.30 \times 2.4 \times 60 = 51.39 \end{aligned}$$

$$\begin{aligned} l=2: & p_1 f_1(\mathbf{x}_0) c(2|1) + p_3 f_3(\mathbf{x}_0) c(2|3) \\ &= 0.05 \times 0.10 \times 10 + 0.30 \times 2.4 \times 50 = 36.05 \end{aligned}$$

$$\begin{aligned} l=3: & p_1 f_1(\mathbf{x}_0) c(3|1) + p_2 f_2(\mathbf{x}_0) c(3|2) \\ &= 0.05 \times 0.10 \times 200 + 0.65 \times 0.63 \times 100 = 41.95 \end{aligned}$$

When $l=2$, minimum value is obtained. So, discriminate $\mathbf{x}_0$ to π_2 .

Exercises:

2. Suppose there are three population groups π_1 , π_2 , π_3 , the pdf of a sample x_0 from them are $f_1(x_0) = 0.1$, $f_2(x_0) = 0.63$, $f_3(x_0) = 2.4$, respectively. The prior probabilities of this sample x_0 from π_1 , π_2 , π_3 are $p_1 = 0.05$, $p_2 = 0.65$, $p_3 = 0.3$, respectively.

- (1) Try to determine which group the sample x_0 should be divided into by maximum posterior probability criterion;
- (2) Try to determine which group the sample x_0 should be divided into by Minimum Expected Cost of Misclassification (ECM) criterion if misclassification cost matrix is known as follows

Discriminate as \ True group	π_1	π_2	π_3
π_1	$C(1 1) = 0$	$C(2 1) = 10$	$C(3 1) = 200$
π_2	$C(1 2) = 20$	$C(2 2) = 0$	$C(3 2) = 100$
π_3	$C(1 3) = 60$	$C(2 3) = 50$	$C(3 3) = 0$

Solution: By maximum posterior probability criterion (5.3.2), we obtain

$$P(\pi_1|x_0) = \frac{p_1 f_1(x_0)}{\sum_{i=1}^3 p_i f_i(x_0)} = 0.004$$

$$P(\pi_2|x_0) = \frac{p_2 f_2(x_0)}{\sum_{i=1}^3 p_i f_i(x_0)} = 0.361$$

$$P(\pi_3|x_0) = \frac{p_3 f_3(x_0)}{\sum_{i=1}^3 p_i f_i(x_0)} = 0.635$$

$$P(\pi_i | \mathbf{x}) = \frac{p_i f_i(\mathbf{x})}{\sum_{i=1}^k p_i f_i(\mathbf{x})}, \quad i = 1, 2, \dots, k \quad (5.3.2)$$

So, x_0 belongs to the third group.

By the minimum ECM criterion (5.3.23), we have

$$l=1 \quad f_2(x_0)p_2C(1|2) + f_3(x_0)p_3C(1|3) = 51.39$$

$$l=2 \quad f_1(x_0)p_2C(2|1) + f_3(x_0)p_3C(2|3) = 36.05$$

$$l=3 \quad f_1(x_0)p_1C(3|1) + f_2(x_0)p_2C(3|2) = 41.95$$

So, x_0 belongs to the second group.

$$\mathbf{x} \in \pi_l, \quad \text{if} \quad \sum_{\substack{j=1 \\ j \neq l}}^k p_j f_j(\mathbf{x}) c(l|j) = \min_{1 \leq i \leq k} \sum_{\substack{j=1 \\ j \neq i}}^k p_j f_j(\mathbf{x}) c(i|j) \quad (5.3.23)$$



Assignment: See the assignment file in BB

